Siqi Yan

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Highlights

- 10 years of experience in C/C++, Python, MATLAB, Linux and ROS development
- Skilled in robotic vision/navigation, IoT and embedded system design, prototyping, and integration across both software and hardware
- Interdisciplinary experience in academic research, algorithm design, PCB soldering, firmware implementation, system deployment and bug diagnosis following first principles
- Experienced with Arduino, Raspberry Pi and NVIDIA Jetson

Education

Studies toward PhD in Transportation Engineering Jan 2022 - Aug 2025
University of Alberta, Edmonton, Canada
- Research focused on pose graph optimization for mapping in dynamic road environments. Designed and implemented graph-based nonlinear optimization pipeline using g2o framework.

MSc in Computing Science Sep 2019 - Jan 2022
University of Alberta, Edmonton, Canada
- Thesis: Improving Automated Vehicle Safety with Collective Perception
- Research focused on multi-view graph-based optimization for simulated scene reconstruction, enabling improved vehicle and pedestrian detection through collective perception.

BSc in Computing Science Sep 2013 - Dec 2018
University of Alberta, Edmonton, Canada

Work History

Graduate Research Assistant Jan 2022 - Aug 2025
University of Alberta, Edmonton, Canada
- Led a research theme focused on developing vision-based detection, tracking and wireless communication systems for Connected Intelligent Transportation Systems (C-ITS) use cases.
- Conducted field tests and system demonstrations.
- Collaborated with a commercial software team on mobile application development.

Software Developer (Part time, Contract) Jun 2023 - Feb 2025
Maparobo Inc., Calgary, Canada (Hybrid)
- Contributed to the development of mobile robotics applications using ROS, Python and C++.
- Designed a simulation-to-physical validation workflow to accelerate development despite limited hardware availability.

- Collaborated with a cross-functional software team to integrate and validate features.

Graduate Research Assistant Jan 2022 - Oct 2022
University of Alberta, Edmonton, Canada
- Developed a large scale road construction zone mapping system using RTAB-Map and stereo cameras.
- Collaborated with a multi-disciplinary team to collect road data.
- Led the design and development of a Python-based data analysis application, enabling non-technical team members to independently process and analyze traffic data, improving workflow efficiency.

Graduate Teaching Assistant Sep 2019 - Jan 2022
University of Alberta, Edmonton, Canada
- Taught CMPUT 412 Experimental Mobile Robotics
- Taught CMPUT 379 Operating Systems

Relevant Experience

Open Source Project: BananaKit2 Development Platform Jan 2025 – Present
- Designed and built a modular embedded systems development platform.
- Implemented the platform firmware using C and C++.
- Project link: https://github.com/siqiyan/BananaKit2

Volunteer: CS Team Developer Jan 2017 - Dec 2019
University of Alberta RoboMasters, Edmonton, Canada
- Designed and assembled multiple fully/semi-automated robots and drones.
- Participated in the ICRA 2018/2019 RoboMaster AI Challenge and received finalist and 3rd prize.
- Trained new members by facilitating weekly workshops.

Publications

Chao Gao, Siqi Yan, Ryan Hayward, and Martin Müller, A Transferable Neural Network for Hex, ICGA Journal 2018 Jun 2018

Awards

IEEE International Conference on Robotics and Automation 2019 Robomaster AI Challenge, 3rd Prize May, 2019
Montreal, Canada

RoboMaster 2018 International Regional Competition 3rd Prize Jul, 2018
Shenzhen, China

IEEE International Conference on Robotics and Automation 2018 DJI Robomaster AI Challenge Finalist May, 2018
Brisbane, Australia